

Introduction to Probability

Preliminaries

We often encounter the use of the term probability in daily life, and it is used in different ways. Some examples:

- i In a cricket match, a fair coin is tossed in the beginning. The probability that this coin will show up heads is $1/2$ and tails is $1/2$. These are however theoretical values of probabilities given a fair coin.
- ii In the case above, it is usually believed that the coin is unbiased. However, we really don't know whether that is the case. To verify this, we need to repeat the experiment (in this case coin tossing experiment) say n a number of times and take the limit of the ratio of successes to n , as n becomes large. The probability values are estimated here through repetition of the experiment.

We call the first kind “Classical probability” (or “Priori probability”)

We call the second kind “Frequentist probability” (or “Posteriori probability”).

The two kinds of perspectives given above are practically very highly restrictive.

Some instances:

The classical as well as frequentist perspectives have limitations when we have an infinite number of possible outcomes. Examples:

- (i) You are picking a real number from the closed interval $[0, \pi]$.
- (ii) You are tossing a coin till head appears.
- (iii) You wish to model a coin that gives heads with probability say $\frac{1}{e}$.

The problem is solved by unifying the two approaches by using the axiomatic definition of probability. The axiomatic approach not only handles both perspectives but also overcomes the limitations.

First, let us solve one probability problem with infinite sample space

A fair coin is tossed until a head turns up for the first time. Find the probability that

- (i) a head turns up for the first time after an even number of tosses.
- (ii) a head turns up for the first time after an odd number of tosses.

Note that the number of tosses need not be finite.

All possible outcomes of the experiment are given by the sample space

$$S = \{1, 2, 3, \dots\}$$

Let E and O denote the events that for the first time a head turns up after an even number of tosses and an odd number of tosses respectively. Then E , O are proper subsets of S and are given by

$$E = \{2, 4, 6, \dots\},$$

$$O = \{1, 3, 5, \dots\}.$$

We have $S = \{1, 2, 3, \dots\}$, $E = \{2, 4, 6, \dots\}$ and $O = \{1, 3, 5, \dots\}$.

$$\begin{aligned}(i) \quad P[E] &= \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots \text{ (a convergent infinite series)} \\ &= \frac{\frac{1}{4}}{1 - \frac{1}{4}} \\ &= \frac{1}{3}\end{aligned}$$

(ii) $E \cup O = S$ and the events E and O are disjoint, that is, $E \cap O = \emptyset$.

$$\text{Hence } P[O] = 1 - \frac{1}{3} = \frac{2}{3}$$

Definition (Axiomatic Definition of Probability)

The set of all possible outcomes is the sample space, S . A subset of S is called an event. Then $P : S \rightarrow [0, 1]$ has to satisfy the following Kolmogorov axioms:

- ① $0 \leq P(A) \leq 1$ for all events A .
- ② $P(S) = 1$.
- ③ For any countable collection of events $A_1, A_2, \dots$ which are disjoint, i.e. $A_i \cap A_j = \emptyset$ for all $i \neq j$, we have

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

Result

$$P(\emptyset) = 0.$$

Proof.

By axiom 1, $P(\emptyset)$ lies in the interval $[0, 1]$. As S and $\emptyset$ are disjoint, by axiom 2 and axiom 3 we have

$$\begin{aligned} 1 = P(S) &= P(S \cup \emptyset \cup \emptyset \dots) = P(S) + P(\emptyset) + P(\emptyset) + \dots \\ &= 1 + P(\emptyset) + P(\emptyset) + \dots \end{aligned}$$

Hence $P(\emptyset) = 0$.



Now, it follows from axiom 3 that if $A_1 \cap A_2 = \emptyset$ then

$$P(A_1 \cup A_2) = P(A_1) + P(A_2).$$

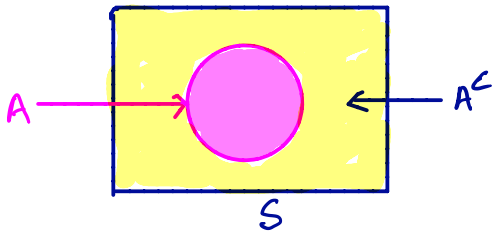
Reason: Choose $A_i = \emptyset$ for $i \geq 3$. Then

$$\begin{aligned} P(A_1 \cup A_2) &= P\left(\bigcup_{i=1}^{\infty} A_i\right) \\ &= \sum_{i=1}^{\infty} P(A_i) \quad (\text{by axiom 3}) \\ &= P(A_1) + P(A_2) + P(\emptyset) + P(\emptyset) + \cdots \\ &= P(A_1) + P(A_2) + 0 + 0 + \cdots \\ &= P(A_1) + P(A_2) \end{aligned}$$

Result

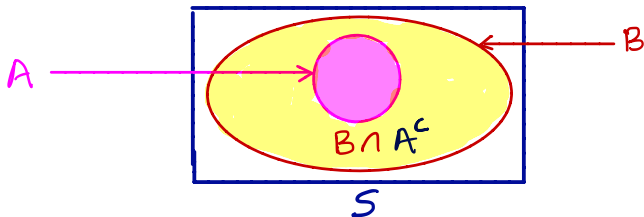
- ① $P(A^c) = 1 - P(A)$.
- ② $A \subseteq B \Rightarrow P(A) \leq P(B)$.
- ③ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

1) $P(A) + P(A^c) = P(S) = 1$ because A and A^c are disjoint. This implies $P(A^c) = 1 - P(A)$.



2) We have $B = A \cup (B \cap A^c)$.

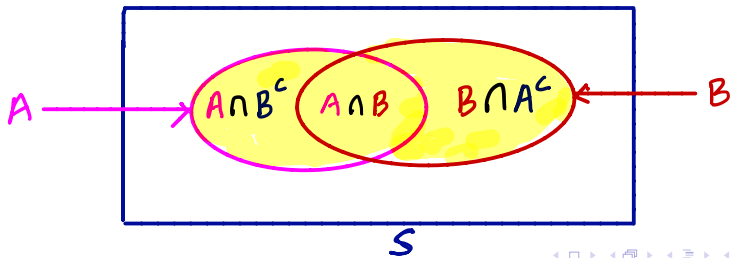
Then $P(B) = P(A) + P(B \cap A^c) \geq P(A)$.



$$3) P(A \cup B) = P(A) + P(B \cap A^c).$$

$$P(B) = P(A \cap B) + P(B \cap A^c).$$

$$\text{Hence } P(A \cup B) = P(A) + P(B) - P(A \cap B).$$



Exercise Problem

Show that

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C).$$

Also, deduce the Boole's inequality:

$$P(A \cup B \cup C) \leq P(A) + P(B) + P(C).$$

Result (Inclusion-Exclusion formula in Probability)

$$\begin{aligned} P\left(\bigcup_i^n A_i\right) &= \sum_{i=1}^n P(A_i) - \sum_{1 \leq i < j \leq n} P(A_i \cap A_j) \\ &+ \sum_{1 \leq i < j < k \leq n} P(A_i \cap A_j \cap A_k) - \cdots + (-1)^{n-1} P(A_1 \cap \cdots \cap A_n). \end{aligned}$$

Exercise: Hat Problem

A group of n people enter a restaurant and give their hats to the hat-keeper. On return, the hat-keeper redistributes the hats back at random.

- (i) What is the probability P_n that no person gets his/her correct hat?
- (ii) What happens to P_n as n tends to infinity? (Answer: $\frac{1}{e}$)

Definition (Independent events)

Two events A and B are independent if

$$P(A \cap B) = P(A)P(B).$$

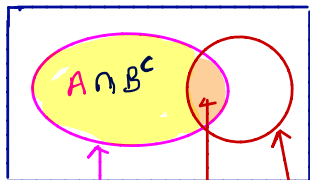
Otherwise, A and B are said to be dependent.

Result

If A and B are independent then A and B^c are independent.

Proof.

$$\begin{aligned} P(A \cap B^c) &= P(A) - P(A \cap B) \\ &= P(A) - P(A)P(B) \\ &= P(A)(1 - P(B)) \\ &= P(A)P(B^c) \end{aligned}$$



A

$A \cap B$

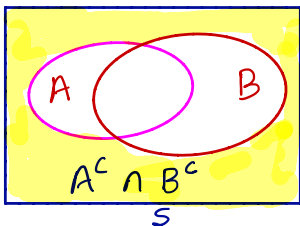
B

Result

If A and B are independent then A^c and B^c are independent.

Proof.

$$\begin{aligned}P(A^c \cap B^c) &= 1 - P(A \cup B) \\&= 1 - (P(A) + P(B) - P(A \cap B)) \\&= 1 - (P(A) + P(B) - P(A)P(B)) \\&= (1 - P(A))(1 - P(B)) \\&= P(A^c)P(B^c)\end{aligned}$$



Example (Birthday problem)

How many people are needed in a room for there to be a probability that two people have the same birthday to be at least 0.5? Ignore leap years and assume that all birthdays are equally likely.

Let us assume that there are k people in the room.

We solve this by finding the probability of no birthday match, $1 - P(k)$. From 365 possible birthdays, the total number of possibilities of birthday combinations is 365^k .

For nobody to have the same birthday, the first person can have any birthday, the second has 364 else to choose, etc. Hence

$$\begin{aligned} P(\text{a shared birthday}) &= 1 - P(\text{no shared birthdays}) \\ &= 1 - \frac{365 \cdot 364 \cdot 363 \cdots (365 - (k - 1))}{365^k} \\ &= 1 - P(k). \end{aligned}$$

We have,

$$1 - P(22) = 0.4757,$$

$$1 - P(23) = 0.5073.$$

The lowest k for which the probability exceeds 0.5 is, $k = 23$.

Answer: 23

Example

A person has n keys in his pocket. He selects one at random once and tries to unlock. What is the possibility that he will succeed at the r th trial if

- (i) the process is done with replacement.*
- (ii) the process is done without replacement.*

With replacement:

The person has to fail in the first $r - 1$ trials and succeed in the r th. So the probability is

$$\frac{(n-1)(n-1)\cdots(n-1)(1)}{n^r} = \frac{(n-1)^{r-1}}{n^r}.$$

Without replacement: The probability is given by

$$\frac{(n-1)(n-2)\cdots(n-r+1)(1)}{n(n-1)\cdots(n-r+1)} = \frac{1}{n}.$$

Conditional Probability

If $P[B] > 0$ then

$$P[A|B] = \frac{P[A \cap B]}{P[B]}.$$

If $P[A] > 0$ then

$$P[B|A] = \frac{P[A \cap B]}{P[A]}.$$

The probability of an event A given that another event B has happened need not be same as the probability of B given that A has happened.

There is a nice relation between the two conditional probabilities, and Bayes' theorem presents it.

Baye's Theorem

Suppose that $B_1, \dots, B_n$ are events from sample space S . Let
 $B_i \neq \emptyset$ for all i ,
 $B_i \cap B_j = \emptyset$ for all $i \neq j$, and
 $\bigcup_{i=1}^n B_i = S$. Then for any event A with $P(A) > 0$,

$$P[B_j|A] = \frac{P[A \cap B_j]}{P[A]} = \frac{P[A|B_j] \cdot P[B_j]}{\sum_{i=1}^n P[A|B_i] \cdot P[B_i]}$$

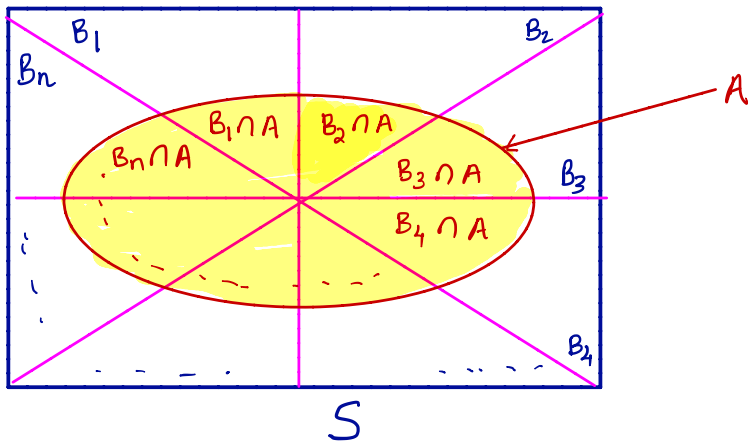
Proof.

$$\text{Clearly, } P[B_j|A] = \frac{P[A \cap B_j]}{P[A]} = \frac{P[A|B_j] \cdot P[B_j]}{P[A]}$$

Now, it is enough to show that

$$P[A] = \sum_{i=1}^n P[A|B_i] \cdot P[B_i]$$

(the statement above is known as the Theorem on Total Probability)



Proof.

continued..

$$\text{We have, } A = A \cap S = A \cap \left(\bigcup_{i=1}^n B_i \right) = \bigcup_{i=1}^n (A \cap B_i)$$

As $(A \cap B_i) \cap (A \cap B_j) = \emptyset$ for all $i \neq j$, we get

$$P[A] = \sum_{i=1}^n P[A \cap B_i]$$

$$= \sum_{i=1}^n P[A|B_i] \cdot P[B_i]$$

$$\text{Thus, } P[B_j|A] = \frac{P[A|B_j] \cdot P[B_j]}{\sum_{i=1}^n P[A|B_i] \cdot P[B_i]}$$



Example

Suppose we have a screening test that tests whether a patient has a particular disease. We denote positive and negative results as positive and negative respectively, and D denotes the person having disease in the population. Suppose that the test is not absolutely accurate, and

$$P(\text{positive} \mid D) = 95\%$$

$$P(\text{positive} \mid D^c) = 8\%$$

$$P(D) = 0.9\%.$$

What is the probability that a person has the disease given that he received a positive result?

$$\begin{aligned}
 P(D \mid \text{positive}) &= \frac{P(\text{positive} \mid D)P(D)}{P(\text{positive} \mid D)P(D) + P(\text{positive} \mid D^c)P(D^c)} \\
 &= \frac{0.95 \cdot (0.009)}{0.95 \cdot (0.009) + 0.08 \cdot (1 - 0.009)} \\
 &= 0.097347
 \end{aligned}$$

Note that the probability that a person has the disease given that he received a positive result is just 9.7347%.

Problem

A MNC has its branches at Ahmadabad, Bengaluru and Chennai wherein they have 1729, 4104 and 7999 employees respectively. For the purpose of dealing with transfer requests from employees, MNC has divided the employees broadly in to two categories Type I and Type II. At Ahmadabad, Bengaluru and Chennai 17%, 2% , 9% of the employees are of Type II (respectively). The company transfers an employee of Type II. What is the probability that the employee worked at Ahmadabad?

Let I be the event that a Type II employee is transferred (from any location).

Let A , B and C denote the event that a randomly selected employee works at Ahmadabad, Bengaluru and Chennai respectively. Then,

$$P[A] = 1729/13382, P[B] = 4104/13382 \text{ and } P[C] = 7999/13382.$$

We have

$$P[I|A] = 17/100, P[I|B] = 2/100, \text{ and } P[I|C] = 9/100.$$

$$P[A|II] = \frac{P(A \cap II)}{P[II]}$$

$$= \frac{P[II|A]P[A]}{P[II|A]P[A] + P[II|B]P[B] + P[II|C]P[C]}$$

$$= \frac{(0.17)(1729/13382)}{(0.17)(1729/13382) + (0.02)(4104/13382) + (0.09)(7999/13382)}$$

$$= 0.2682$$

Problem

A bag contains 90 fair coins ($P(H) = \frac{1}{2} = P(T)$) and 10 unfair coins which flip with $P(H) = \frac{3}{4}$, $P(T) = \frac{1}{4}$. A coin is picked at random and tossed n times and each one of the n tosses were heads.

- (i) What is the probability that the picked coin is the unfair coin?
- (ii) Find the least value of n that gives probability that the picked coin is the unfair is at least 90%?

Let A denote the event that n tosses of the coin gave n heads.

Let B_1 denote the event that the coin is unfair and B_2 denote the event that the coin is fair.

$$\begin{aligned}(i) P[B_1|A] &= \frac{P[A|B_1] \cdot P[B_1]}{P[A|B_1] \cdot P[B_1] + P[A|B_2] \cdot P[B_2]} \\ &= \frac{\left(\frac{3}{4}\right)^n \cdot (0.1)}{\left(\frac{3}{4}\right)^n \cdot (0.1) + \left(\frac{1}{2}\right)^n \cdot (0.9)}\end{aligned}$$

(ii) When $n = 10$, we get $P[B_1|A] = 0.864997$

When $n = 11$, we get $P[B_1|A] = 0.905757$

Hence the least value of n is 11.